

Advanced Materials Modelling

Details of Hartree-Fock approximation

Kohn-Sham DFT formulation

Iterative techniques

DFT treatment of spin

Sergey Levchenko

*Skolkovo Institute of Science and Technology
Moscow, Russia*

Ground-state electronic structure problem

$$\left[-\frac{1}{2} \sum_i \frac{\partial^2}{\partial \mathbf{r}_i^2} - \sum_i \sum_J \frac{Z_J}{|\mathbf{r}_i - \mathbf{R}_J|} + \sum_{i>j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + V^{\text{ext}}(\{\mathbf{r}\}) \right] \Psi(\{\mathbf{r}\}, \{\sigma\}) = E \Psi(\{\mathbf{r}\}, \{\sigma\})$$

$\Psi(\{\mathbf{r}\}, \{\sigma\})$ – many-body wave function, depends on spatial ($\mathbf{r}_i$) and spin (σ_i) coordinates of particles (also on nuclear coordinates ($\mathbf{R}_J$) and $V^{\text{ext}}(\{\mathbf{r}\})$)

Ground-state electronic structure problem

$$\left[-\frac{1}{2} \sum_i \frac{\partial^2}{\partial \mathbf{r}_i^2} - \sum_i \sum_J \frac{Z_J}{|\mathbf{r}_i - \mathbf{R}_J|} + \sum_{i>j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + V^{\text{ext}}(\{\mathbf{r}\}) \right] \Psi(\{\mathbf{r}\}, \{\sigma\}) = E \Psi(\{\mathbf{r}\}, \{\sigma\})$$

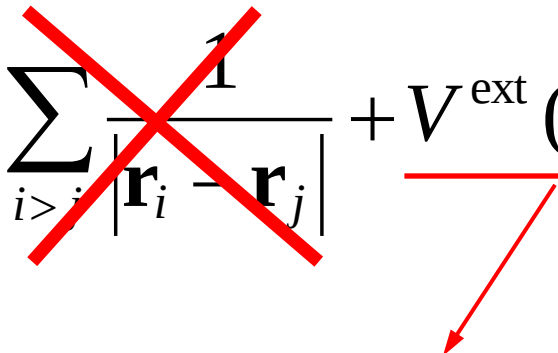
$\Psi(\{\mathbf{r}\}, \{\sigma\})$ – many-body wave function, depends on spatial ($\mathbf{r}_i$) and spin (σ_i) coordinates of particles (also on nuclear coordinates ($\mathbf{R}_J$) and $V^{\text{ext}}(\{\mathbf{r}\})$)

- already include approximations (Born-Oppenheimer, non-relativistic, no spin-orbit coupling, no magnetic field)
- wave function depends on $4N$ variables (spatial + spin)
- electrons interact via Coulomb forces

Ground-state electronic structure problem

$$\hat{H} = -\frac{1}{2} \sum_i \frac{\partial^2}{\partial \mathbf{r}_i^2} - \sum_i \sum_J \frac{Z_J}{|\mathbf{r}_i - \mathbf{R}_J|} + \sum_{i>j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + V^{\text{ext}}(\{\mathbf{r}\})$$

Ground-state electronic structure problem

$$\hat{H} = -\frac{1}{2} \sum_i \frac{\partial^2}{\partial \mathbf{r}_i^2} - \sum_i \sum_J \frac{Z_J}{|\mathbf{r}_i - \mathbf{R}_J|} + \sum_{i>j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{V^{\text{ext}}(\{\mathbf{r}\})}{}$$


$$V^{\text{ext}}(\{\mathbf{r}\}) = \sum_i v(\mathbf{r}_i)$$



$$\hat{H}_1 = \sum_i \hat{h}_i, \hat{h}_i \psi_i = \varepsilon_i \psi_i$$

$$\Psi(\{\mathbf{r}\}) = \psi_1(\mathbf{r}_1) \psi_2(\mathbf{r}_2) \dots \psi_N(\mathbf{r}_N) = \prod_i \psi_i(\mathbf{r}_i)$$

$$\hat{H}_1 \Psi(\{\mathbf{r}\}) = \left(\sum_i \varepsilon_i \right) \Psi(\{\mathbf{r}\})$$

Variational principle

$$E = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad \langle \Psi | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\} = 1$$

$$\langle \Psi | \hat{H} | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \hat{H} \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\}$$

Variational principle

$$\hat{H}\Psi = E\Psi$$



$$E = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad \langle \Psi | \Psi \rangle \equiv \int \Psi^*({\bf r}, \{\sigma\}) \Psi({\bf r}, \{\sigma\}) d{\bf r} d\{\sigma\} = 1$$

$$\langle \Psi | \hat{H} | \Psi \rangle \equiv \int \Psi^*({\bf r}, \{\sigma\}) \hat{H} \Psi({\bf r}, \{\sigma\}) d{\bf r} d\{\sigma\}$$

**Observation: minimization of E with respect to $\Psi \rightarrow$
Schrödinger equation**

$$\delta E = \frac{\delta E}{\delta \Psi^*} \delta \Psi^* + \frac{\delta E}{\delta \Psi} \delta \Psi = 0 \rightarrow \frac{\delta E}{\delta \Psi^*} = \frac{\delta E}{\delta \Psi} = 0$$

because $\delta \Psi^*$ and $\delta \Psi$ are *arbitrary* (complex conjugate of the same equation, consider only one)

Variational principle

$$\hat{H}\Psi = E\Psi$$



$$E = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad \langle \Psi | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\} = 1$$

$$\langle \Psi | \hat{H} | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \hat{H} \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\}$$

**Observation: minimization of E with respect to $\Psi \rightarrow$
Schrödinger equation**

$$\delta \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\langle \delta \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} - \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle^2} \langle \delta \Psi | \Psi \rangle = 0$$

$$\langle \delta \Psi | \hat{H} - \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} | \Psi \rangle = 0 \rightarrow \hat{H}\Psi - E\Psi = 0$$

Variational principle

$$\hat{H}\Psi = E\Psi$$



$$E = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad \langle \Psi | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\} = 1$$

$$\langle \Psi | \hat{H} | \Psi \rangle \equiv \int \Psi^*(\{\mathbf{r}\}, \{\sigma\}) \hat{H} \Psi(\{\mathbf{r}\}, \{\sigma\}) d\{\mathbf{r}\} d\{\sigma\}$$

Another way -- minimization with constraints:

$$L = \langle \Psi | \hat{H} | \Psi \rangle - E(\langle \Psi | \Psi \rangle - 1)$$

Lagrangian

**constraint
(normalization)**

$$\delta L = \langle \delta \Psi | \hat{H} | \Psi \rangle - E \langle \delta \Psi | \Psi \rangle = 0 \rightarrow (\hat{H} - E)\Psi = 0$$

Hartree approximation

Hartree approximation

Apply variational principle:

$$\frac{\delta}{\delta \psi_k^*} \left(\langle \Psi | \hat{H} | \Psi \rangle - \sum_i \varepsilon_i (\langle \psi_i | \psi_i \rangle - 1) \right) = 0$$

$$\langle \Psi | \hat{H} | \Psi \rangle = \sum_i \langle \psi_i | \hat{h} | \psi_i \rangle + \sum_{i>j} \int \frac{|\psi_i(\mathbf{r}_i)|^2 |\psi_j(\mathbf{r}_j)|^2}{|\mathbf{r}_i - \mathbf{r}_j|} d^3 r_i d^3 r_j$$



$$\left(\hat{h} + \sum_{j \neq k} \int \frac{|\psi_j(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \right) \psi_k(\mathbf{r}) = \varepsilon_k \psi_k(\mathbf{r})$$

Ground-state electronic structure problem

$$\hat{H}_1 = \sum_i \hat{h}_i, \hat{h}_i \psi_i = \varepsilon_i \psi_i$$

$$\hat{H}_1 \Psi(\{\mathbf{r}\}) = \left(\sum_i \varepsilon_i \right) \Psi(\{\mathbf{r}\})$$

$$\Psi(\{\mathbf{r}\}) = \prod_i \psi_i(\mathbf{r}_i) \text{ – eigenfunction of } \hat{H}_1$$

Ground-state electronic structure problem

$$\hat{H}_1 = \sum_i \hat{h}_i, \hat{h}_i \psi_i = \varepsilon_i \psi_i$$

$$\hat{H}_1 \Psi(\{\mathbf{r}\}) = \left(\sum_i \varepsilon_i \right) \Psi(\{\mathbf{r}\})$$

$$\Psi(\{\mathbf{r}\}) = \prod_i \psi_i(\mathbf{r}_i) \text{ – eigenfunction of } \hat{H}_1$$

However,

$$\tilde{\Psi}(\{\mathbf{r}\}) = \psi_1(\mathbf{r}_1) \dots \psi_n(\mathbf{r}_{n+1}) \psi_{n+1}(\mathbf{r}_n) \dots \psi_N(\mathbf{r}_N)$$

is also a solution of $\hat{H}_1 \Psi(\{\mathbf{r}\}) = E \Psi(\{\mathbf{r}\})$ with exactly the same energy

Ground-state electronic structure problem

$$\hat{H}_1 = \sum_i \hat{h}_i, \hat{h}_i \psi_i = \varepsilon_i \psi_i$$

$$\Psi(\{\mathbf{r}\}, \{\sigma\}) = \sum_{\mathbf{q}} C_{\mathbf{q}} \hat{P}_{\mathbf{q}} \left[\prod_i \psi_i(\mathbf{r}_{q_i}) s_i(\sigma_{q_i}) \right]$$

index-permutation operator

$$\hat{H}_1 \Psi(\{\mathbf{r}\}, \{\sigma\}) = \left(\sum_i \varepsilon_i \right) \Psi(\{\mathbf{r}\}, \{\sigma\})$$

In general, the coefficients $C_{\mathbf{q}}$ are almost arbitrary (apart from normalization) ... but not for electrons!

Fermions versus bosons

Indistinguishable particles



Permutation of two particles cannot change any observable



Wavefunction can change only by a phase factor



$$\Psi(\mathbf{r}_1, \mathbf{r}_2) \rightarrow e^{i\phi} \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

Permuting again should change the wavefunction back

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) \rightarrow e^{i\phi} \Psi(\mathbf{r}_2, \mathbf{r}_1) \rightarrow e^{2i\phi} \Psi(\mathbf{r}_1, \mathbf{r}_2) \Rightarrow e^{2i\phi} = 1$$

Fermions versus bosons

Indistinguishable particles



Permutation of two particles cannot change any observable



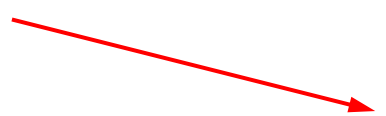
Wavefunction can change only by a phase factor



$$\Psi(\mathbf{r}_1, \mathbf{r}_2) \rightarrow e^{i\phi} \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

Permuting again should change the wavefunction back


$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1)$$


$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

Fermions versus bosons

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1)$$

fermions:

cannot occupy the same
quantum state

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

bosons:

can occupy the same
quantum state

Fermions versus bosons

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1)$$

fermions:

cannot occupy the same
quantum state

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

bosons:

can occupy the same
quantum state

Spin-statistics theorem: spin-1/2 particles are all fermions, integer-spin – bosons (from relativity)



electrons are fermions

Many-electron wave function

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1)$$

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

fermions:

cannot occupy the same
quantum state

bosons:

can occupy the same
quantum state

electrons are fermions

$$\Psi(\{\mathbf{r}\}, \{\sigma\}) = \sum_{\mathbf{q}} C_{\mathbf{q}} \hat{P}_{\mathbf{q}} \left[\prod_i \psi_i(\mathbf{r}_{q_i}) s_i(\sigma_{q_i}) \right]$$

$$C_{\mathbf{q}} = \frac{(-1)^{n(\mathbf{q})}}{\sqrt{N!}}$$

smallest number of permutations to
revert back to original order

normalization factor

Many-electron wave function

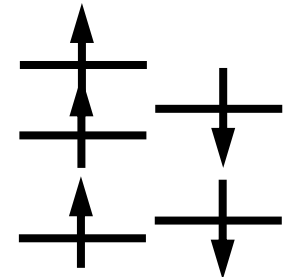
$$\Psi(\{\mathbf{r}\}, \{\sigma\}) = \frac{1}{\sqrt{N!}} \sum_{\mathbf{q}} (-1)^{n(\mathbf{q})} \hat{P}_{\mathbf{q}} \left[\prod_i \psi_i(\mathbf{r}_{q_i}) s_i(\sigma_{q_i}) \right]$$



$$\Psi = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(r_1) s_1(\sigma_1) & \psi_2(r_1) s_2(\sigma_1) & \dots & \psi_N(r_1) s_N(\sigma_1) \\ \psi_1(r_2) s_1(\sigma_2) & \psi_2(r_2) s_2(\sigma_2) & \dots & \psi_N(r_2) s_N(\sigma_2) \\ \vdots & \vdots & \dots & \vdots \\ \psi_1(r_N) s_1(\sigma_N) & \psi_2(r_N) s_2(\sigma_N) & \dots & \psi_N(r_N) s_N(\sigma_N) \end{bmatrix}$$

Slater determinant

$$\left(\sum_i \hat{h}_i \right) \Psi = E \Psi, \quad \hat{h}_i \psi_i = \varepsilon_i \psi_i$$



Hartree-Fock approximation

$$\hat{H} = \sum_i \hat{h}(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

$$\langle \Psi | \hat{H} | \Psi \rangle = \sum_j h_{jj} + \frac{1}{2} \sum_{k \neq l} \left(\langle kl | kl \rangle - \langle s_k | s_l \rangle \langle kl | lk \rangle \right)$$

$$\langle kl | kl \rangle - \langle s_k | s_l \rangle \langle kl | lk \rangle = 0 \text{ when } k = l$$



self-interaction free!

$$\langle \Psi | \hat{H} | \Psi \rangle = \sum_j h_{jj} + \frac{1}{2} \sum_{k,l} \left(\langle kl | kl \rangle - \langle s_k | s_l \rangle \langle kl | lk \rangle \right)$$

Hartree-Fock approximation

$$\frac{\delta}{\delta \psi_k^*} \left(\langle \Psi | \hat{H} | \Psi \rangle - \sum_i \varepsilon_{ij} (\langle \psi_i s_i | \psi_j s_j \rangle - \delta_{ij}) \right) = 0$$



$$\underbrace{\left[\hat{h} + \sum_j (\hat{J}_j - \hat{K}_j) \right]}_{\text{Fock operator}} \psi_k = \sum_j \varepsilon_{kj} \psi_j$$

??

Hartree-Fock approximation

$$\frac{\delta}{\delta \psi_k^*} \left(\langle \Psi | \hat{H} | \Psi \rangle - \sum_i \varepsilon_{ij} (\langle \psi_i s_i | \psi_j s_j \rangle - \delta_{ij}) \right) = 0$$



$$\underbrace{\left[\hat{h} + \sum_j (\hat{J}_j - \hat{K}_j) \right]}_{\text{Fock operator}} \psi_k = \sum_j \varepsilon_{kj} \psi_j$$

??

The Fock operator is *invariant* under any *unitary* transformation

Hartree-Fock approximation

$$\left[\hat{h} + \sum_j (\hat{J}_j - \hat{K}_j) \right] \psi_k = \sum_j \varepsilon_{kj} \psi_j$$

$$\psi'_k = \sum_j \psi_j U_{jk}, \quad \mathbf{U}^T \mathbf{U} = \delta$$

$$\varepsilon'_{ki} = \sum_{mn} U_{mk}^* \varepsilon_{mn} U_{ni} = \mathbf{U}^T \boldsymbol{\varepsilon} \mathbf{U}$$

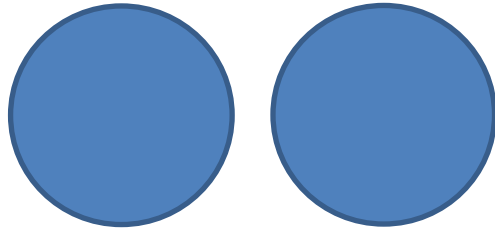
$\boldsymbol{\varepsilon}^T = \boldsymbol{\varepsilon} \rightarrow$ **always can find $\mathbf{U}$ such that $\boldsymbol{\varepsilon}'$ is diagonal:**

$$\left[\hat{h} + \sum_j (\hat{J}_j - \hat{K}_j) \right] \psi_k = \varepsilon_k \psi_k$$

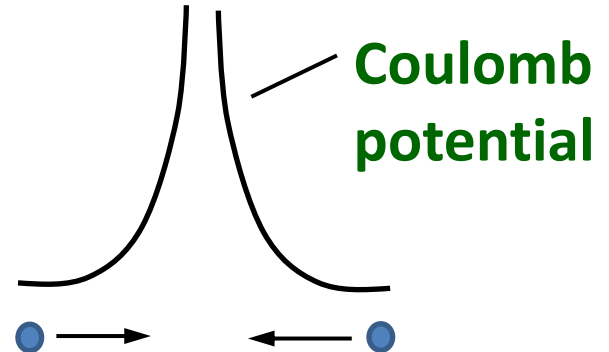
canonical Hartree-Fock equations

Errors in the Hartree-Fock approximation

Dynamic correlation:

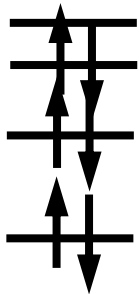


versus

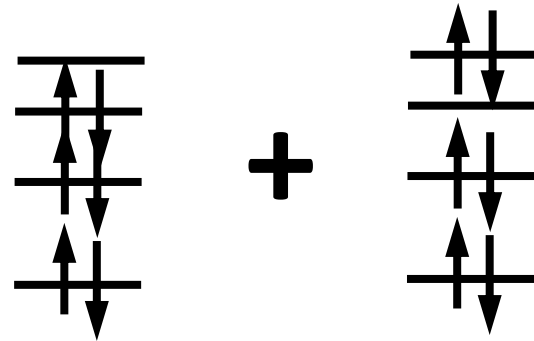


smaller e density, large $\partial\psi / \partial\mathbf{r}$

Non-dynamic (static) correlation:



versus



(quasi)degenerate HOMO-LUMO

HF approximation $\rightarrow$ $\geq 90\%$ of total energy, overestimates ionicity

Density functional theory

Nobel lecture ...

$$\Phi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

- **The many-body wavefunction of 1000 electrons cannot be stored in the whole universe**

Van Vleck catastrophe (1936)

W. Kohn, Rev. Mod. Phys. 11, 1253 (1999).

C. Some meta-physical-chemical considerations

The following remarks are related to a very old paper by one of my teachers, J. H. Van Vleck (1936), in which he discusses a problem with many-body wave functions, later referred to as the Van Vleck catastrophe.

I begin with a provocative statement. *In general the many-electron wave function $\Psi(r_1, \dots, r_N)$ for a system of N electrons is not a legitimate scientific concept, when $N \geq N_0$, where $N_0 \approx 10^3$.*

A simpler example

□ Ne atom

$$\Phi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \mathbf{r}_4, \mathbf{r}_5, \mathbf{r}_6, \mathbf{r}_7, \mathbf{r}_8, \mathbf{r}_9, \mathbf{r}_{10})$$

10 electrons

30 coordinates

10 entries for each coordinate **10^{30} entries**

1 byte per entry **10^{30} bytes**

128 GB per smartphone **$\sim 10^{19}$ smartphones**

7×10^9 people on earth **1.5×10^9 smartphones / person**

□ **Storing is not an option (maybe calculating on demand is)**

Density-functional theory

- ❑ In principle, we need to know the many-electron wave function

$$\Phi(\{\mathbf{r}_i\}) = \Phi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$$

depending on the coordinates of all electrons

- ❑ According to Hohenberg and Kohn, only the electron density is required

$$n(\mathbf{r})$$

depending on three coordinates: x, y, z

Questions

- ❑ Do we really need the many-body wave function?
- ❑ Can we compute the electron distribution without explicitly knowing the WF?

$$n(\mathbf{r}) = \left\langle \Phi(\{\mathbf{r}_i\}) \left| \sum_i^N \delta(\mathbf{r} - \mathbf{r}_i) \right| \Phi(\{\mathbf{r}_i\}) \right\rangle$$

$$n(\mathbf{r}) = N \int \Phi^*(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N) \Phi(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N) d^3r_2 \dots d^3r_N$$

Hohenberg-Kohn theorem

P. Hohenberg, W. Kohn, Phys. Rev. 136 , B864 (1964)

- **I: The total energy of an interacting system of electrons is a unique functional of the density**

$$E = F[n(\mathbf{r})] + \int V_{\text{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}$$

The functional F does not depend on the external potential!

- **II: The energy takes its minimum at the ground-state density**

Proof of HK theorem

- **I: Uniqueness:** We need to show that there is only one potential that leads to a certain density

Given $H^e, V_{ext}, E = \langle \Phi(\{\mathbf{r}_i\}) | H^e | \Phi(\{\mathbf{r}_i\}) \rangle, n(\mathbf{r})$

Assume there is another potential such that we have

$$\tilde{H}^e, \tilde{V}_{ext}, \tilde{E} = \langle \tilde{\Phi}(\{\mathbf{r}_i\}) | \tilde{H}^e | \tilde{\Phi}(\{\mathbf{r}_i\}) \rangle, \tilde{n}(\mathbf{r}) = n(\mathbf{r})$$

$$\tilde{E} = \langle \tilde{\Phi} | \tilde{H}^e | \tilde{\Phi} \rangle < \langle \Phi | \tilde{H}^e | \Phi \rangle = \langle \Phi | H^e - V_{ext} + \tilde{V}_{ext} | \Phi \rangle$$

$$E = \langle \Phi | H^e | \Phi \rangle < \langle \tilde{\Phi} | H^e | \tilde{\Phi} \rangle = \langle \tilde{\Phi} | \tilde{H}^e - \tilde{V}_{ext} + V_{ext} | \tilde{\Phi} \rangle$$

$$\tilde{E} < E + \langle \Phi | \tilde{V}_{ext} - V_{ext} | \Phi \rangle$$

$$E < \tilde{E} - \langle \tilde{\Phi} | \tilde{V}_{ext} - V_{ext} | \tilde{\Phi} \rangle$$

Proof of HK theorem

$$+ \quad \tilde{E} < E + \langle \Phi | \tilde{V}_{\text{ext}} - V_{\text{ext}} | \Phi \rangle = E + \int d\mathbf{r} (\tilde{V}_{\text{ext}} - V_{\text{ext}}) n(\mathbf{r})$$

$$E < \tilde{E} + \langle \tilde{\Phi} | V_{\text{ext}} - \tilde{V}_{\text{ext}} | \tilde{\Phi} \rangle = \tilde{E} + \int d\mathbf{r} (V_{\text{ext}} - \tilde{V}_{\text{ext}}) \tilde{n}(\mathbf{r})$$

$$\updownarrow \tilde{n}(\mathbf{r}) = n(\mathbf{r})$$

$$\tilde{E} + \int d\mathbf{r} (V_{\text{ext}} - \tilde{V}_{\text{ext}}) n(\mathbf{r})$$

$$\tilde{E} + E < E + \tilde{E}$$

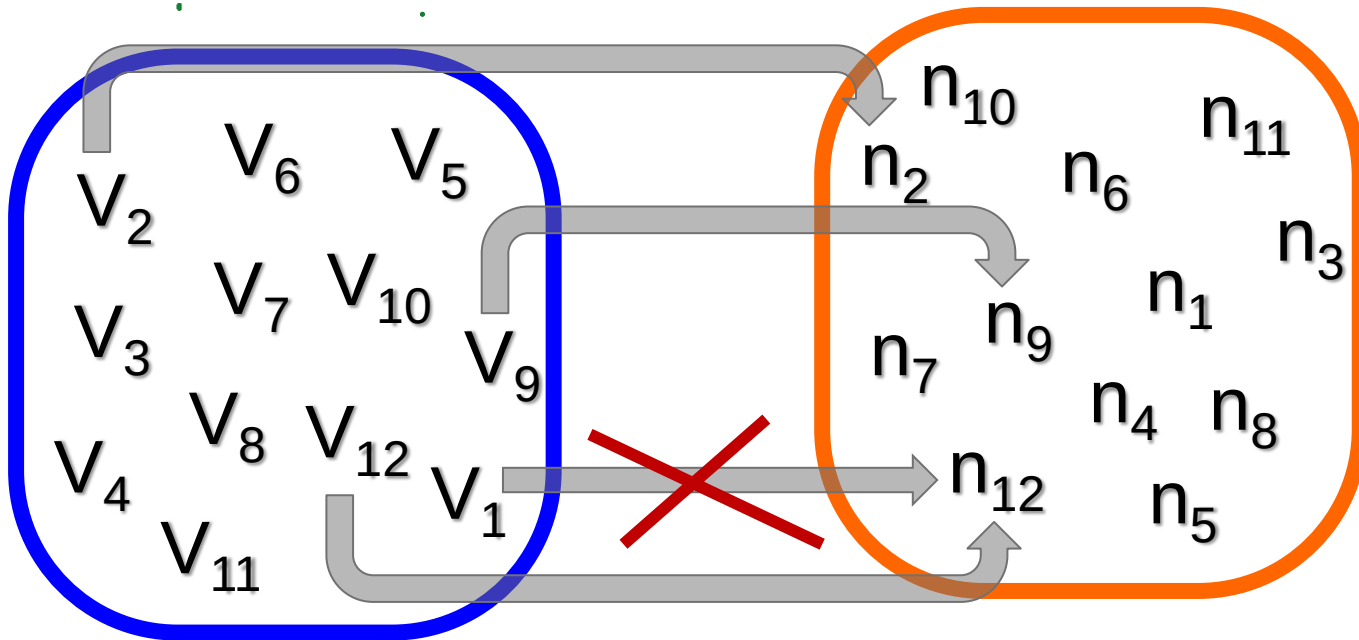
impossible!



$$\tilde{n}(\mathbf{r}) \neq n(\mathbf{r})$$

Proof of HK theorem

- There is a one-to-one correspondence between density and potential



Proof of HK theorem

- **Part II: The ground state energy can be obtained variationally: The density that minimizes the total energy is the ground-state density**

Proof of HK theorem

- **Part II: The ground state energy can be obtained variationally: The density that minimizes the total energy is the ground-state density**

$$n(\mathbf{r}) \rightarrow \hat{V}_{ext} \rightarrow \hat{H}^e = \hat{F} + \hat{V}_{ext} \rightarrow \Phi(\{\mathbf{r}_i\}) \Rightarrow \langle \Phi | \hat{F} | \Phi \rangle \equiv F[n(\mathbf{r})]$$

The total energy is a functional of the density:

$$E[n(\mathbf{r})] = F[n(\mathbf{r})] + \int V_{ext}(\mathbf{r})n(\mathbf{r})d\mathbf{r}$$

A density that is a ground state of some external potential is called *V*-representable

$$E[n'(\mathbf{r})] = F[n'(\mathbf{r})] + \int V_{ext}(\mathbf{r})n'(\mathbf{r})d\mathbf{r} = \langle \Phi' | \hat{F} | \Phi' \rangle + \langle \Phi' | \hat{V}_{ext} | \Phi' \rangle$$

Proof of HK theorem

- **Part II: The ground state energy can be obtained variationally: The density that minimizes the total energy is the ground-state density**

$$E[n'(\mathbf{r})] = F[n'(\mathbf{r})] + \int V_{ext}(\mathbf{r})n'(\mathbf{r})d\mathbf{r} = \langle \Phi' | \hat{F} | \Phi' \rangle + \langle \Phi' | \hat{V}_{ext} | \Phi' \rangle$$

For a ground state density:

$$E[n(\mathbf{r})] = F[n(\mathbf{r})] + \int V_{ext}(\mathbf{r})n(\mathbf{r})d\mathbf{r} = \langle \Phi | \hat{F} | \Phi \rangle + \langle \Phi | \hat{V}_{ext} | \Phi \rangle$$

From variational principle:

$$\langle \Phi' | \hat{F} | \Phi' \rangle + \langle \Phi' | \hat{V}_{ext} | \Phi' \rangle > \langle \Phi | \hat{F} | \Phi \rangle + \langle \Phi | \hat{V}_{ext} | \Phi \rangle$$

$$E[n'(\mathbf{r})] > E[n(\mathbf{r})]$$

Kohn-Sham theory

- ❑ The HK theorem is exact for the ground state
- ❑ It is beautiful as we only need to know one = *THE* functional to describe any system
Atoms, molecules, solids, surfaces, nano-structures
- ❑ The only problem is that we don't know this functional
Thus DFT is not practical so far
- ❑ Kohn and Sham set the stage for the practicality of DFT
W. Kohn, L. Sham, Phys. Rev. A 140,1133 (1965)

Kohn-Sham theory

- One can divide the energy functional into three contributions:

$$E = F[n(\mathbf{r})] + \int V_{\text{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r}$$

$$F[n(\mathbf{r})] = T[n(\mathbf{r})] + \frac{1}{2} \iint \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' + E_{\text{xc}}[n(\mathbf{r})]$$

- Even if we don't know the expressions for the kinetic energy and the exchange-correlation energy, we can formally vary it with respect to the density constraining the particle number N :

$$\frac{\delta(E - \mu N)}{\delta n(\mathbf{r})} = 0 = \frac{\delta T[n(\mathbf{r})]}{\delta n(\mathbf{r})} + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{\text{xc}}[n(\mathbf{r})]}{\delta n(\mathbf{r})} + V_{\text{ext}}(\mathbf{r}) - \mu$$

$$L = E - \mu N, \quad N = \int n(\mathbf{r})d^3r$$

Kohn-Sham theory

$$\frac{\delta(E - \mu N)}{\delta n(\mathbf{r})} = 0 = \frac{\delta T[n(\mathbf{r})]}{\delta n(\mathbf{r})} + \left[\int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{xc}[n(\mathbf{r})]}{\delta n(\mathbf{r})} + V_{ext}(\mathbf{r}) \right] - \mu$$

- **Introducing the exchange-correlation (xc) potential**

$$V_{xc}(\mathbf{r}) = \frac{\delta E_{xc}[n(\mathbf{r})]}{\delta n(\mathbf{r})}$$

- **and the effective potential**

$$V_{eff}(\mathbf{r}) = \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + V_{xc}(\mathbf{r}) + V_{ext}(\mathbf{r})$$

- **we arrive at**

$$\frac{\delta T[n(\mathbf{r})]}{\delta n(\mathbf{r})} + V_{eff}(\mathbf{r}) - \mu = 0$$

Kohn-Sham theory

- Since this equation looks like an equation for non-interacting particles, there must be single-particle functions solving it.

$$\frac{\delta T[n(\mathbf{r})]}{\delta n(\mathbf{r})} + V_{\text{eff}}(\mathbf{r}) - \mu = 0$$

- We still don't have a good expression for the kinetic energy

This problem is, however, diminished by introducing single-particle functions which give the exact density:

$$n(\mathbf{r}) = \sum_i^N |u_i(\mathbf{r})|^2$$

$$L = E - \mu N, N = \int n(\mathbf{r}) d^3r \rightarrow L = E - \sum_i \epsilon_i (\int u_i^*(\mathbf{r}) u_i(\mathbf{r}) d^3r - 1)$$

Kohn-Sham theory

- We will now carry out the variation with respect to the density through a variation with respect to the single-particle functions.

$$\begin{aligned}
 L = & -\frac{1}{2} \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) \nabla^2 u_i(\mathbf{r}) \\
 & + \int d\mathbf{r} \quad n(\mathbf{r}) \quad V_{\text{ext}}(\mathbf{r}) \\
 & + \frac{1}{2} \int d\mathbf{r} \quad n(\mathbf{r}) \quad \int d\mathbf{r}' \quad n(\mathbf{r}') \quad \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\
 & + \int d\mathbf{r} \quad n(\mathbf{r}) \quad \epsilon_{\text{xc}}(\mathbf{r}) \\
 & - \sum_i \epsilon_i \left(\int u_i^*(\mathbf{r}) u_i(\mathbf{r}) d^3r - 1 \right)
 \end{aligned}$$

kinetic energy of non-interacting electrons

Kohn-Sham theory

$$\begin{aligned} L = & -\frac{1}{2} \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) \nabla^2 u_i(\mathbf{r}) \\ & + \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) u_i(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) \\ & + \frac{1}{2} \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) u_i(\mathbf{r}) \int d\mathbf{r}' \sum_j u_j^*(\mathbf{r}') u_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\ & + \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) u_i(\mathbf{r}) \epsilon_{\text{xc}}(\mathbf{r}) \\ & - \int d\mathbf{r} \sum_i u_i^*(\mathbf{r}) u_i(\mathbf{r}) \epsilon_i \end{aligned}$$

Kohn-Sham theory

$$\begin{aligned}\delta L = \int d\mathbf{r} \sum_i \delta u_i^*(\mathbf{r}) \bigg\{ & -\frac{1}{2} \nabla^2 u_i(\mathbf{r}) \\ & + u_i(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) \\ & + u_i(\mathbf{r}) \int d\mathbf{r}' \sum_j u_j^*(\mathbf{r}') u_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\ & + u_i(\mathbf{r}) V_{\text{xc}}(\mathbf{r}) \\ & - u_i(\mathbf{r}) \epsilon_i \bigg\}\end{aligned}$$

Kohn-Sham theory

$$\delta L = \int d\mathbf{r} \sum_i \delta u_i^*(\mathbf{r}) \left\{ -\frac{1}{2} \nabla^2 u_i(\mathbf{r}) + u_i(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) + u_i(\mathbf{r}) \int d\mathbf{r}' \sum_j u_j^*(\mathbf{r}') u_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} + u_i(\mathbf{r}) V_{\text{xc}}(\mathbf{r}) - u_i(\mathbf{r}) \epsilon_i \right\}$$

0 fulfilled for each i

$$\left\{ -\frac{1}{2} \nabla^2 + V_{\text{ext}}(\mathbf{r}) + \int d\mathbf{r}' \sum_j u_j^*(\mathbf{r}') u_j(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}(\mathbf{r}) - \epsilon_i \right\} u_i(\mathbf{r}) = 0$$

Kohn-Sham equations

$$\left\{ -\frac{1}{2}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + \int d\mathbf{r}' \left[\sum_j u_j^*(\mathbf{r}') u_j(\mathbf{r}') \right] \frac{1}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}(\mathbf{r}) - \epsilon_i \right\} u_i(\mathbf{r}) = 0$$

$$\left\{ -\frac{1}{2}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + \int d\mathbf{r}' n(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}(\mathbf{r}) - \epsilon_i \right\} u_i(\mathbf{r}) = 0$$

$$\left\{ -\frac{1}{2}\nabla^2 + V_{\text{ext}}(\mathbf{r}) + \int d\mathbf{r}' n(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{xc}}[n(\mathbf{r})] - \epsilon_i \right\} u_i(\mathbf{r}) = 0$$

approximation needed!
contains xc effects and corrections to kinetic energy

Kohn-Sham theory

- ❑ **Caution: The Kohn-Sham eigenvalues were derived as Lagrange parameters of the variation procedure and cannot *a priori* be interpreted as single-particle energies.**
- ❑ **The Kohn-Sham orbitals are auxiliary quantities which produce the exact density.**
- ❑ **The only approximation required for the ground state is V_{xc} .**
- ❑ **The potential is a functional of the density which is obtained from the KS orbitals; these, in turn, give the density. Thus the KS equation must be solved self-consistently.**

Exchange-correlation functionals

□ Local-density approximation (LDA)

xc effects taken from homogenous electron gas

$$E_{xc}^{LDA} = \int d\mathbf{r} \, n(\mathbf{r}) \epsilon_{xc}^{unif} [n(\mathbf{r})]$$

Exchange is known analytically:

$$\epsilon_x(r_s) = -\frac{3}{2\pi} \left(\frac{9\pi}{4} \right)^{\frac{1}{3}} \frac{1}{r_s} \quad r_s = \left[\frac{3}{4\pi n(\mathbf{r})} \right]^{\frac{1}{3}}$$

Correlation: high- and low-density limits are known analytically:

$$\epsilon_c = A \ln(r_s) + B + r_s(C \ln(r_s) + D) \quad - \text{high density}$$

Exchange-correlation functionals

□ Local-density approximation (LDA)

xc effects taken from homogenous electron gas

$$E_{xc}^{LDA} = \int d\mathbf{r} \, n(\mathbf{r}) \epsilon_{xc}^{unif} [n(\mathbf{r})]$$

Exchange is known analytically:

$$\epsilon_x(r_s) = -\frac{3}{2\pi} \left(\frac{9\pi}{4} \right)^{\frac{1}{3}} \frac{1}{r_s} \quad r_s = \left[\frac{3}{4\pi n(\mathbf{r})} \right]^{\frac{1}{3}}$$

Correlation: high- and low-density limits are known analytically:

$$\epsilon_c = -\frac{0.438}{r_s} + \frac{1.33}{r_s^{3/2}} - \frac{1.47}{r_s^2}$$

- low-density (electrons form a BCC lattice called Wigner crystal)

Exchange-correlation functionals

□ Local-density approximation (LDA)

Neither the high nor the low-density limits are useful for ordinary matter

For the intermediate densities, one can parametrize an exact numerical solution (e.g., obtained with quantum Monte Carlo)

e.g., Hedin & Lundquist, 1971

$$\epsilon_c[n(\mathbf{r})] = 0.045 \left[\left(1 + x^3\right) \ln \left(1 + \frac{1}{x}\right) + \frac{x}{2} - x^2 - \frac{1}{3} \right] \quad x = \frac{r_s}{21}$$

Exchange-correlation functionals

- Many features of the exact $E_{xc}[n]$ are known → constraints on the approximate functional

Limiting behavior for $n(\mathbf{r}) \rightarrow \infty$ and $n(\mathbf{r}) \rightarrow 0$

Lieb-Oxford bound:

$$\left\langle \Phi \left| \frac{1}{2} \sum_{i \neq j} \frac{1}{r_{ij}} \right| \Phi \right\rangle - \frac{1}{2} \int \frac{n_{\Phi}(\mathbf{r}) n_{\Phi}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \geq 1.68 \int n_{\Phi}(\mathbf{r})^{\frac{4}{3}} d\mathbf{r}$$

- LDA satisfies these constraints

Exchange-correlation functionals

- Many features of the exact $E_{xc}[n]$ are known → constraints on the approximate functional

Express $E_{xc}[n]$ through *exchange-correlation hole* (change due to XC of probability to find an electron at r when another electron is at r' , can be measured!):

Exchange-correlation functionals

- Many features of the exact $E_{XC}[n]$ are known → constraints on the approximate functional

Express $E_{XC}[n]$ through *exchange-correlation hole* (change due to XC of probability to find an electron at r when another electron is at $r' = r + u$):

$$E_{XC}[n] = \frac{1}{2} \int d\mathbf{r} n(\mathbf{r}) \int d\mathbf{u} \frac{h_{XC}(\mathbf{r}, \mathbf{r} + \mathbf{u})}{|\mathbf{u}|}$$

Sum rules:

$$\int d\mathbf{u} h_{XC}(\mathbf{r}, \mathbf{r} + \mathbf{u}) = -1$$

$$\int d\mathbf{u} h_X(\mathbf{r}, \mathbf{r} + \mathbf{u}) = -1, \quad \int d\mathbf{u} h_C(\mathbf{r}, \mathbf{r} + \mathbf{u}) = 0$$

Local conditions:

$$h_X(\mathbf{r}, \mathbf{r} + \mathbf{u}) \leq 0 \quad \forall \mathbf{r}, \mathbf{u}, \quad h_X(\mathbf{r}, \mathbf{r}) = -\frac{n(\mathbf{r})}{2} \quad (\text{no spin})$$

- LDA satisfies all these constraints!

Exchange-correlation functionals

- ❑ **What about spin?**

Exchange-correlation functionals

□ Local spin density approximation (LSDA)

The total energy is a unique functional of the density, but the wave function is not; for example, any normalized linear combination of wave functions with the same energy gives the same density

If $\hat{H}$ does not depend on spin, there are $2S + 1$ wave functions with the same energy and the same total spin S

Constrain number of spin-up and spin-down electrons separately $\rightarrow E[n_\alpha, n_\beta] = \langle \Phi | \hat{H} | \Phi \rangle_{\Phi \rightarrow n_\alpha, n_\beta}$

Generalizations of the HK theorem to $E[n_\alpha, n_\beta]$ exist

Example: Cu

VOLUME 81, NUMBER 22

PHYSICAL REVIEW LETTERS

30 NOVEMBER 1998

Absolute Band Mapping by Combined Angle-Dependent Very-Low-Energy Electron Diffraction and Photoemission: Application to Cu

V. N. Strocov,^{1,2,*} R. Claessen,¹ G. Nicolay,¹ S. Hüfner,¹ A. Kimura,³ A. Harasawa,³ S. Shin,³ A. Kakizaki,⁴
P. O. Nilsson,² H. I. Starnberg,² and P. Blaha⁵

¹*Fachrichtung Experimentalphysik, Universität des Saarlandes, D-66041 Saarbrücken, Germany*

²*Department of Physics, Chalmers University of Technology and Göteborg University, SE-41296 Göteborg, Sweden*

³*Institute for Solid State Physics, University of Tokyo, Roppongi, Minato-ku, Tokyo 106-8666, Japan*

⁴*Institute of Materials Structure Science, High Energy Accelerator Research Organization (KEK), Ibaraki 305-0801, Japan*

⁵*Institut für Technische Elektrochemie, Technische Universität Wien, A-1060 Wien, Austria*

(Received 18 August 1998)

We present an experimental method to determine the electronic $E(\mathbf{k})$ band structure in crystalline solids absolutely, i.e., with complete control of the three-dimensional wave vector $\mathbf{k}$. Angle-dependent very-low-energy electron diffraction is first applied to determine the unoccupied states whose $\mathbf{k}$ is located on a high-symmetry line parallel to the surface. Photoemission via these states, employing the constant-final-state mode, is then utilized to map the valence bands along this line. We demonstrate the method by application to Cu, and find significant deviation from free-electron-like behavior in the unoccupied states, and from density-functional theory in the occupied states. [S0031-9007(98)07792-8]

Example: Cu

PHYSICAL REVIEW B, VOLUME 63, 205108

Three-dimensional band mapping by angle-dependent very-low-energy electron diffraction and photoemission: Methodology and application to Cu

V. N. Strocov* and R. Claessen

Experimentalphysik II, Universität Augsburg, D-86135 Augsburg, Germany

G. Nicolay and S. Hüfner

Fachrichtung Experimentalphysik, Universität des Saarlandes, D-66041 Saarbrücken, Germany

A. Kimura, A. Harasawa, and S. Shin

Institute for Solid State Physics, University of Tokyo, Roppongi, Minato-ku, Tokyo 106-8666, Japan

A. Kakizaki

Institute of Materials Structure Science, High Energy Accelerator Research Organization (KEK), Ibaraki 305-0801, Japan

H. I. Starnberg and P. O. Nilsson

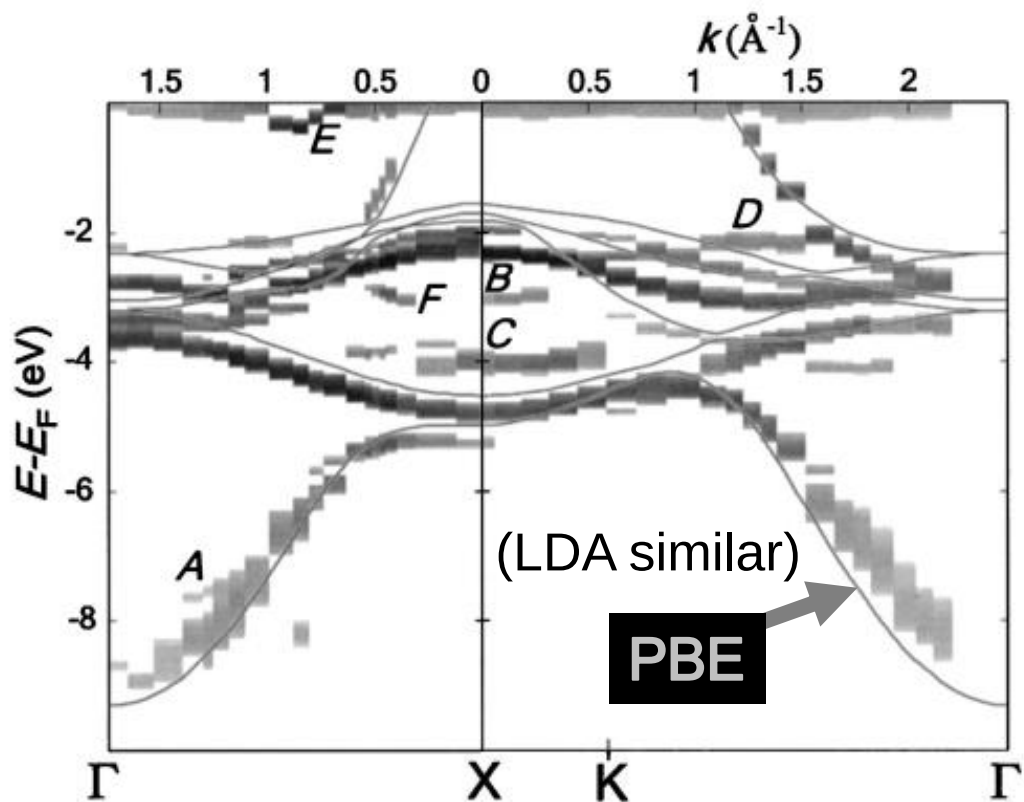
Department of Physics, Chalmers University of Technology and Göteborg University, SE-41296 Göteborg, Sweden

P. Blaha

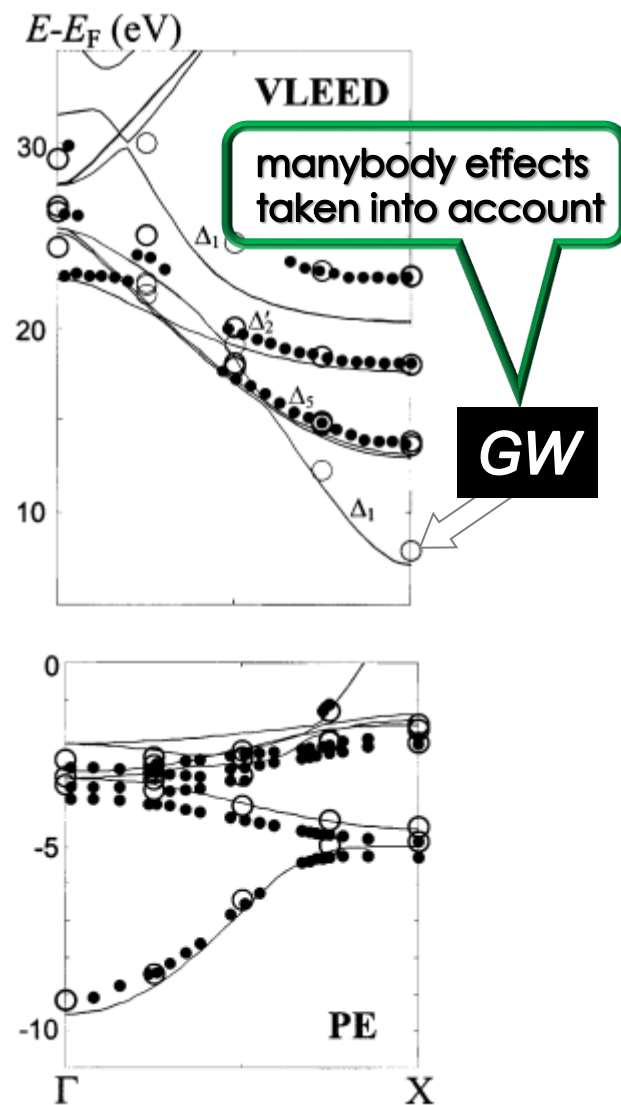
Institut für Physikalische und Theoretische Chemie, Technische Universität Wien, A-1060 Wien, Austria

(Received 17 November 2000; revised manuscript received 2 February 2001; published 27 April 2001)

Example: Cu



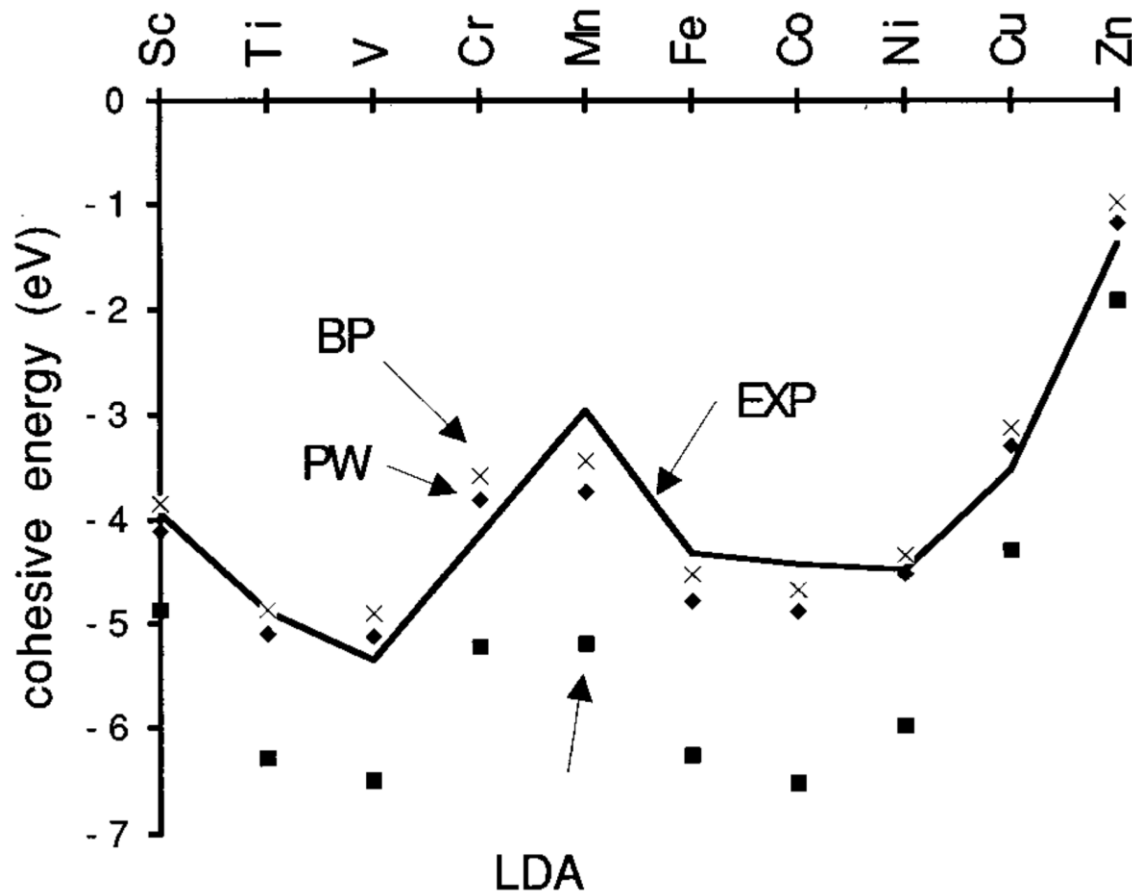
V. N. Strocov, et al.,
Phys. Rev. B 63, 205108 (2001).



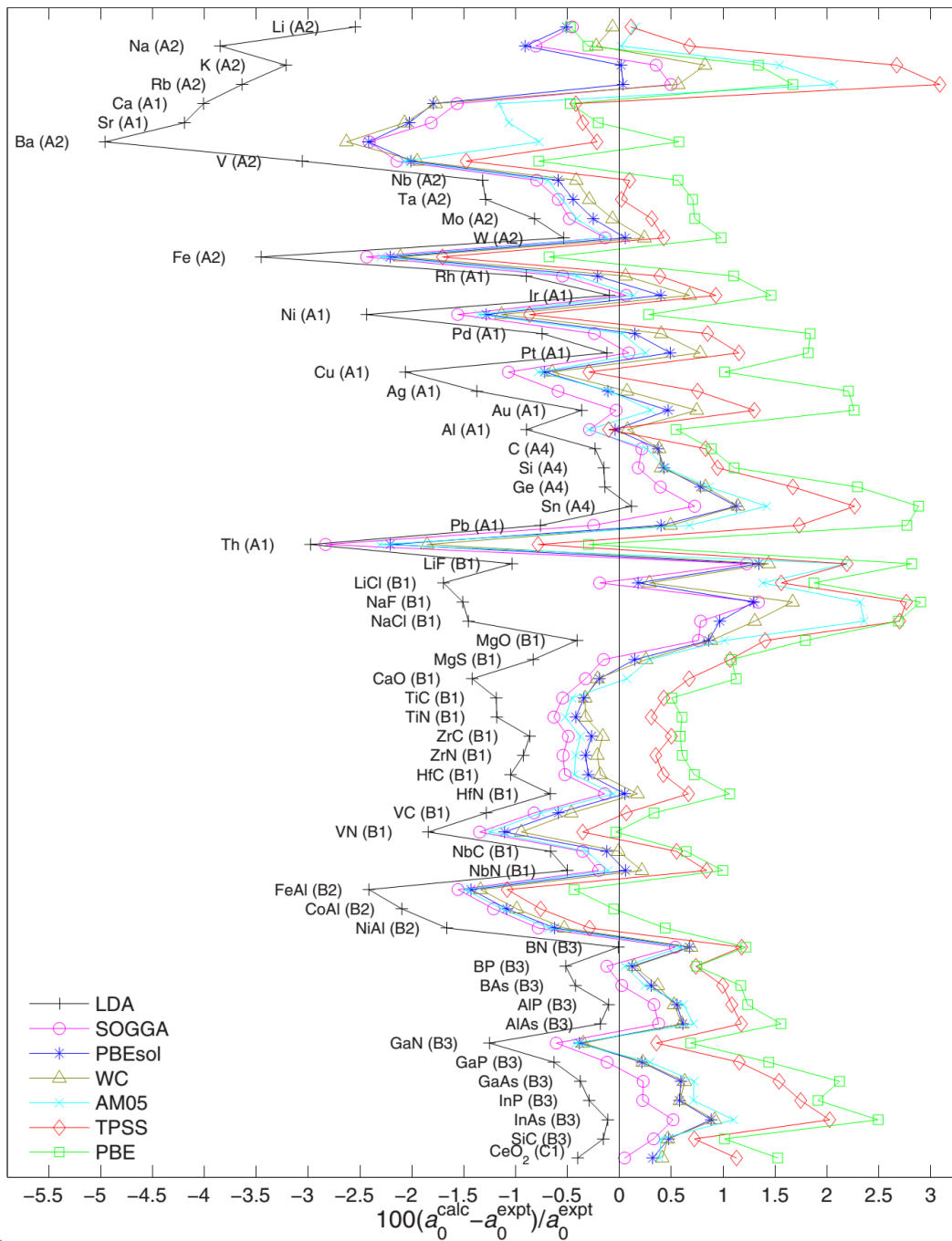
V. N. Strocov, et al.,
Surface Rev. and Lett. 9, 1281 (2002).

Exchange-correlation functionals

□ How well (or badly) does LDA work?



Philipsen and Baerends, Phys. Rev. B, 54, p. 5326 (1996)



Equilibrium lattice constant compared to experiment

P. Haas, F. Tran, and P. Blaha,
Phys. Rev. B 79, 085104 (2009)

Exchange-correlation functionals

□ How well (or badly) does LDA work?

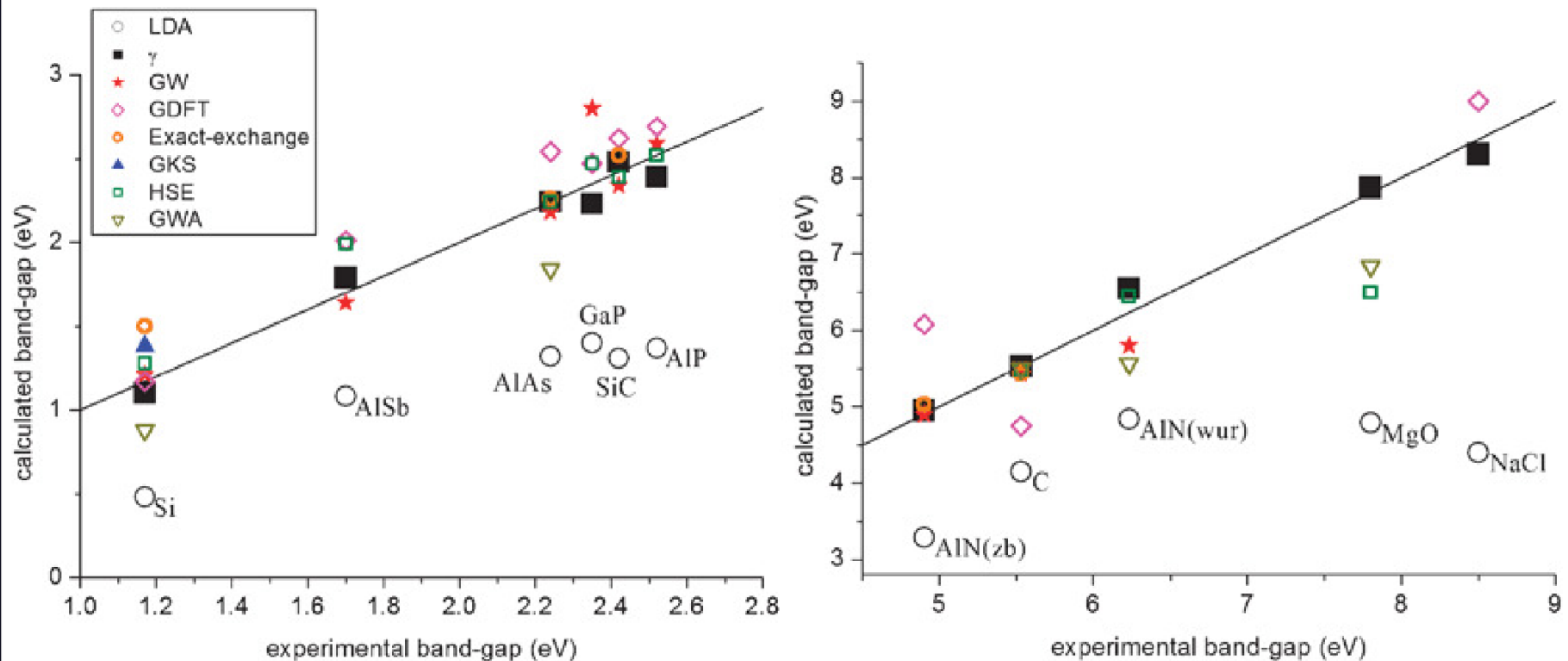
For total energies, E_x is underestimated by about 10%, E_c is overestimated by about 200%, so E_{xc} is good to about 7% (mysterious cancellation of errors).

For bond dissociation energies, LDA overbinds by about 1 eV/bond (30 kcal/mol), so no good for thermochemistry.

Typical bond lengths are underestimated by 1% (unless involving an H atom), so excellent geometries and vibrations. So still used for structure.

Exchange-correlation functionals

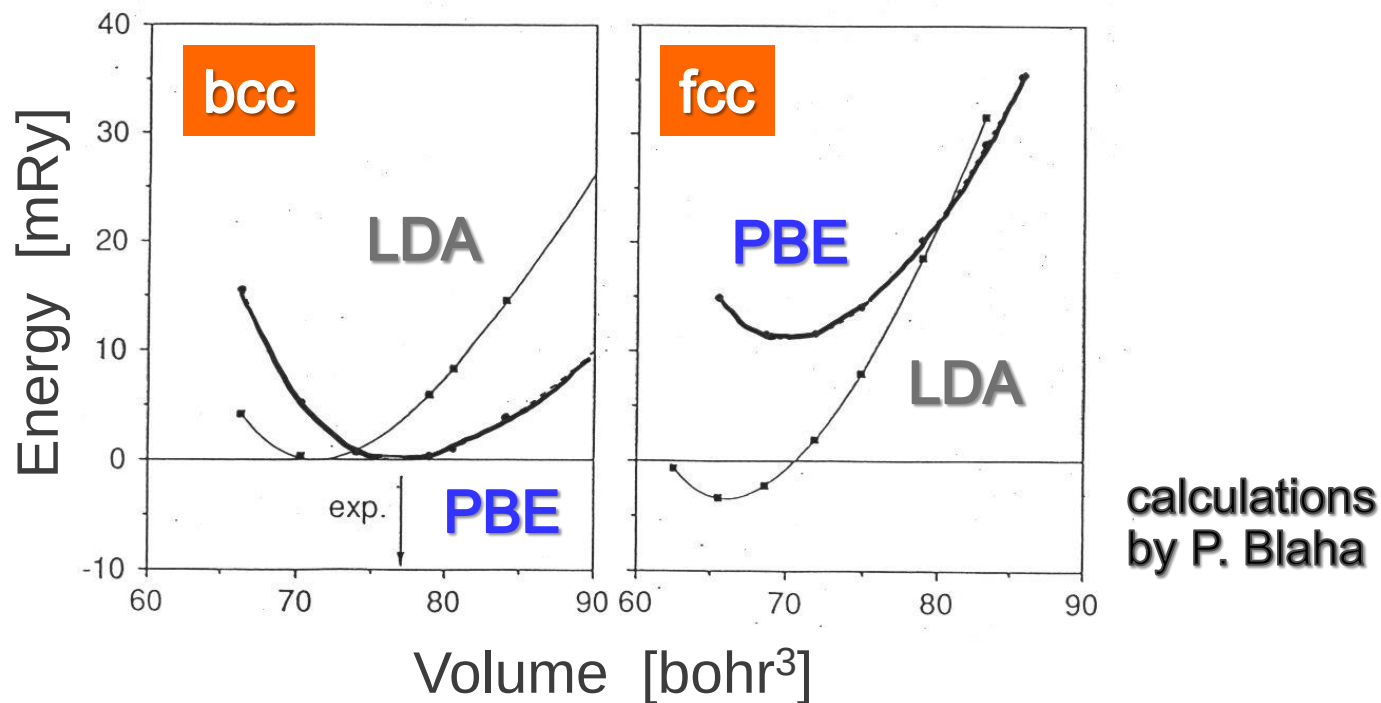
□ How well (or badly) does LDA work?



Philipsen and Baerends, Phys. Rev. B, 54, p. 5326 (1996)

Example: Fe

- It took until 1996 that the proper crystallographic phase of Fe could be correctly obtained by DFT



J. Perdew, K. Burke, and M. Ernzerhof, Phys. Rev. Lett. 77, 3865 (1996).
More than 168 000 citations

Exchange-correlation functionals

□ Generalized gradient approximation (GGA)

Take into account local variations of the density:

$$E_{XC}^{GGA} = \int d\mathbf{r} n(\mathbf{r}) (\epsilon_X^{GGA}[n, |\nabla n|] + \epsilon_C^{GGA}[n, |\nabla n|])$$

Gradient expansion approximation: expand E_{XC} for slowly varying electron gas in terms of ∇n -- slow convergence

Exchange-correlation functionals

□ Generalized gradient approximation (GGA)

Take into account local variations of the density:

$$E_{XC}^{GGA} = \int d\mathbf{r} n(\mathbf{r}) (\epsilon_X^{GGA}[n, |\nabla n|] + \epsilon_C^{GGA}[n, |\nabla n|])$$

Gradient expansion approximation: expand E_{XC} for slowly varying electron gas in terms of ∇n -- slow convergence

$$\epsilon_X^{GGA} = \epsilon_X^{HES}[n] F_X[n, \nabla n] \quad (\text{enhancement factor})$$

$$\text{PBE:} \quad F_X = 1 + \kappa - \frac{\kappa}{1 + \mu s^2 / \kappa}, \quad s = \frac{|\nabla n(\mathbf{r})|}{2\kappa n(\mathbf{r})}$$

$$\epsilon_C^{GGA} = \epsilon_C^{LDA} + \frac{\beta^2}{2\alpha} \ln \left(1 + \frac{2\alpha}{\beta} \frac{s^2 + As^4}{1 + As^2 + A^2s^4} \right)$$

Exchange-correlation functionals

□ Generalized gradient approximation (GGA)

Take into account local variations of the density:

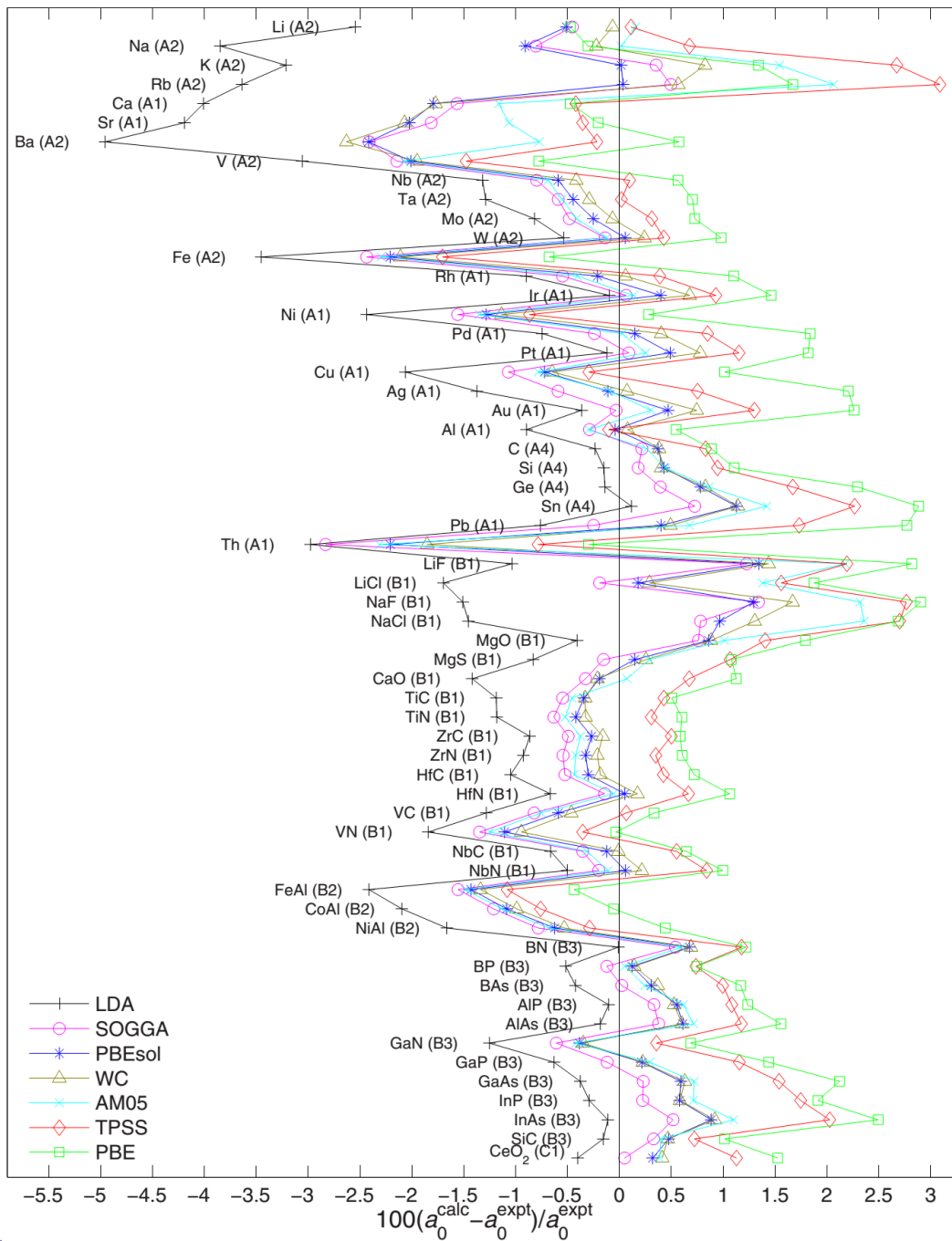
$$E_{XC}^{GGA} = \int d\mathbf{r} n(\mathbf{r}) (\epsilon_X^{GGA}[n, |\nabla n|] + \epsilon_C^{GGA}[n, |\nabla n|])$$

PBE: $\epsilon_X^{GGA} = \epsilon_X^{HES}[n] F_X[n, \nabla n]$ (enhancement factor)

$$F_X = 1 + \kappa - \frac{\kappa}{1 + \mu s^2 / \kappa}, \quad s = \frac{|\nabla n(\mathbf{r})|}{2\kappa n(\mathbf{r})}$$

$$\epsilon_C^{GGA} = \epsilon_C^{LDA} + \frac{\beta^2}{2\alpha} \ln \left(1 + \frac{2\alpha}{\beta} \frac{s^2 + As^4}{1 + As^2 + A^2s^4} \right)$$

- The parameters are chosen to satisfy the constraints (PBE: exchange hole sum rule and local, and Lieb-Oxford bound)

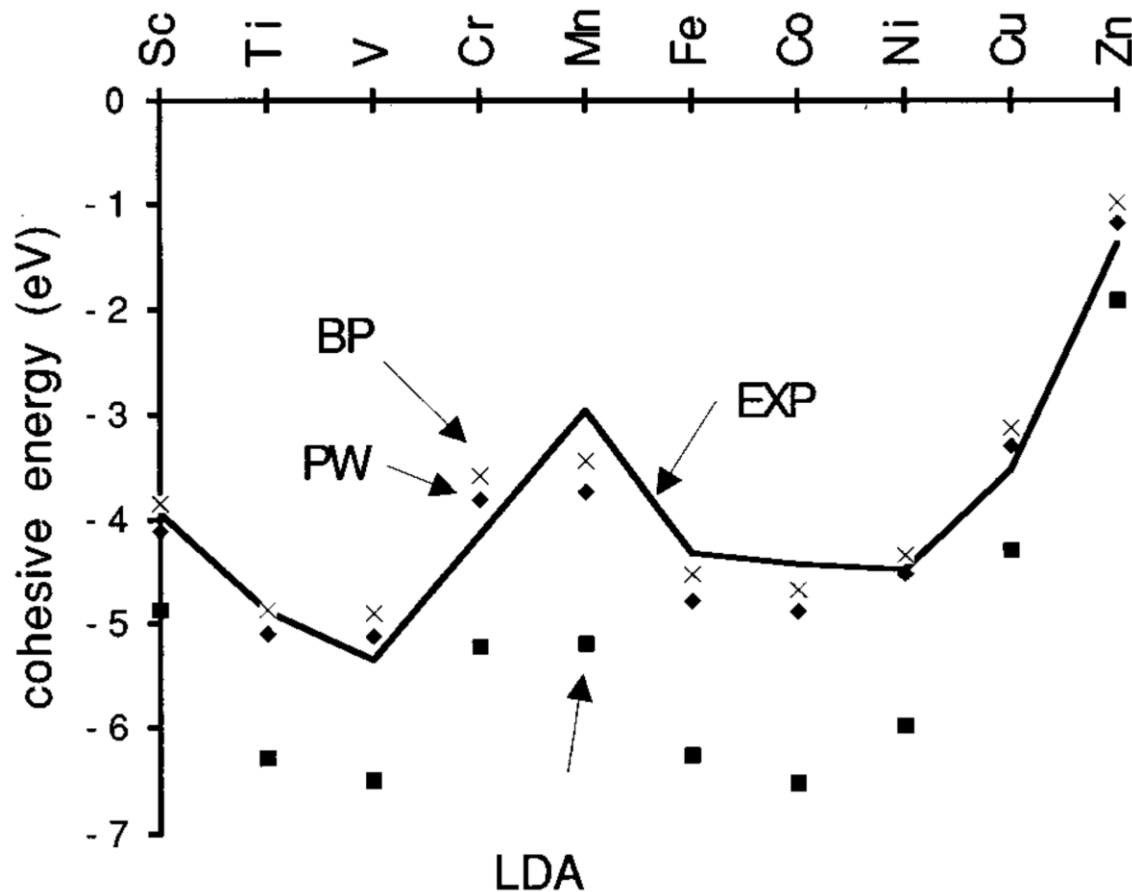


Equilibrium lattice constant compared to experiment

P. Haas, F. Tran, and P. Blaha,
Phys. Rev. B 79, 085104 (2009)

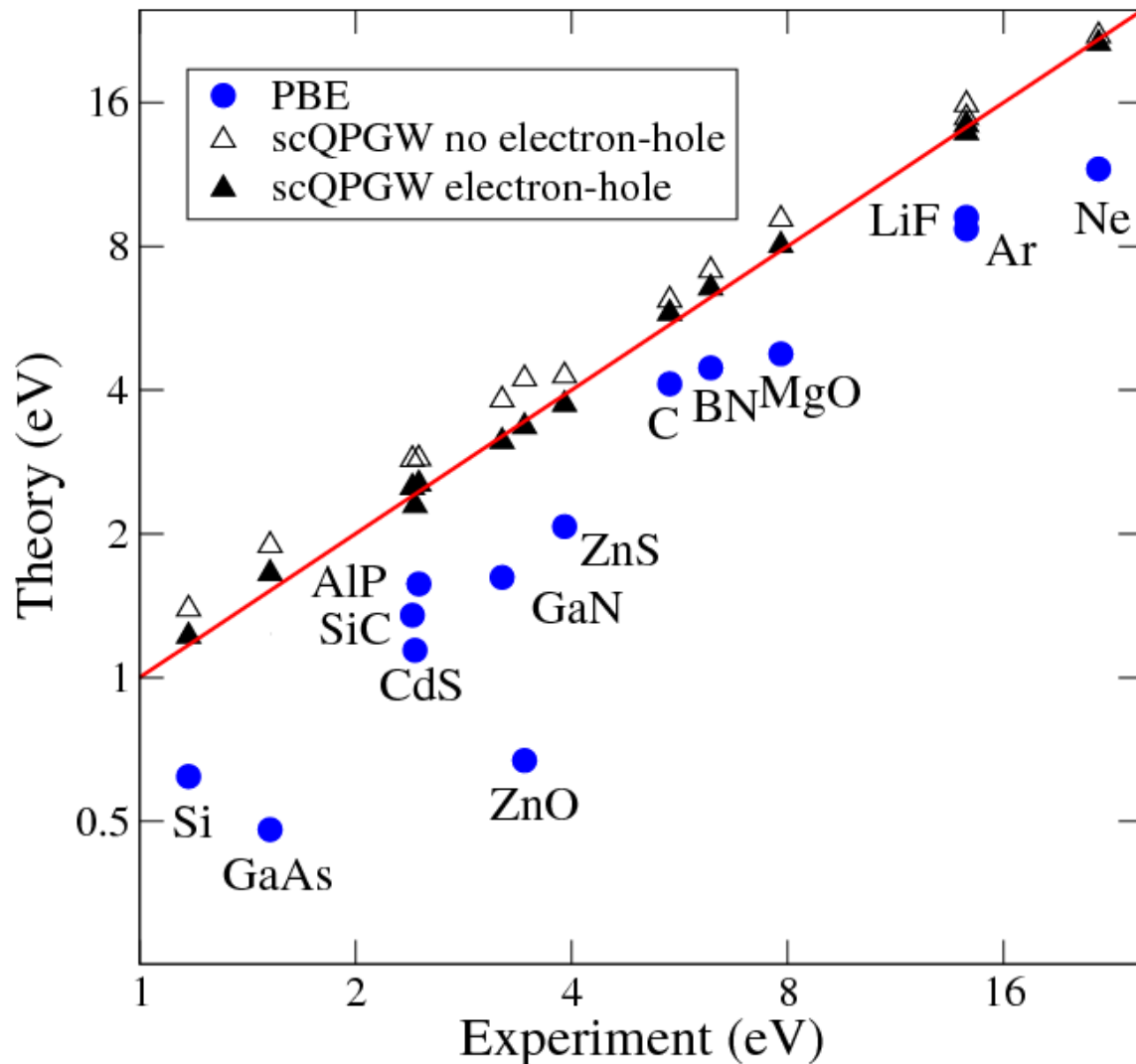
Exchange-correlation functionals

□ Performance of GGAs



Philipsen and Baerends, Phys. Rev. B, 54, p. 5326 (1996)

Exchange-correlation functionals



M. Shishkin, M. Marsman, Phys. Rev. Lett. 95, 246403 (2007)

Exchange-correlation functionals

□ Meta-GGAs

$$E_{XC}^{MGGA} = \int d\mathbf{r} n(\mathbf{r}) \epsilon_{XC}^{MGGA}(n, \nabla n, \tau)$$

with *kinetic energy density* $\tau(\mathbf{r}) = \sum_i^{occup} \frac{1}{2} |\nabla \psi_i(\mathbf{r})|^2$

Although τ is a non-local functional of n , it is available in standard calculations

For a 1-electron system, τ is known exactly (Weizsäcker):

$$\tau^W = \frac{|\nabla n|^2}{8n}$$

Gradient expansion motivates use of τ

Functional is constructed similar to GGAs

Exchange-correlation functionals

□ Meta-GGAs

$$E_{XC}^{MGGA} = \int d\mathbf{r} n(\mathbf{r}) \epsilon_{XC}^{MGGA}(n, \nabla n, \tau)$$

Histogram of lattice constant errors for a set of semiconductors (units: 0.01 Å)

